THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · NCERT-TYPE CHAPTER TEST SET 1

Chapter 10: **How Quantities Combine (Understanding Data)** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The average marks of 27 students is 57 and of another 39 students is 35. The average of all 66 students is: 1

(a) 43 **(b)** 44 **(c)** 46 **(d)** 45

2. The average of 14 numbers is 31. When a number 61 is included, the average of all 15 numbers becomes: 1

(a) 33 **(b)** 34 **(c)** 32 **(d)** 35

3. Marks of 70, 65 and 75 are given weights 6, 6 and 3 respectively. The weighted average is: 1

(a) 68 **(b)** 69 **(c)** 70 **(d)** 71

4. 60 mL of a 15% sugar solution is mixed with 120 mL of a 33% sugar solution. The concentration of the mixture is: 1

(a) 29% **(b)** 26% **(c)** 28% **(d)** 27%

5. In a 100% stacked bar of a household's monthly water use, appliances take 17% of 900 units. The units used by the appliances are: 1

(a) 153 **(b)** 155 **(c)** 154 **(d)** 152

6. The weighted average of values x_1, x_2 with weights w_1, w_2 is: 1

(a) $\dfrac{w_1 + w_2}{x_1 + x_2}$ **(b)** $w_1x_1 + w_2x_2$ **(c)** $\dfrac{x_1 + x_2}{w_1 + w_2}$ **(d)** $\dfrac{w_1x_1 + w_2x_2}{w_1 + w_2}$

7. The average of 30 numbers is 54. The average of all 45 numbers (including 15 more numbers) is 46. The average of the 15 additional numbers is: 1

(a) 31 **(b)** 29 **(c)** 32 **(d)** 30

8. The average of 8 numbers is 32. If the number 39 is removed, the average of the remaining 7 numbers is: 1

(a) 32 **(b)** 33 **(c)** 31 **(d)** 30

9. The average marks of 35 students is 65 and of another 55 students is 47. The average of all 90 students is: 1

(a) 53 **(b)** 55 **(c)** 54 **(d)** 56

10. The average of 21 numbers is 31. When a number 53 is included, the average of all 22 numbers becomes: 1

(a) 31 **(b)** 33 **(c)** 32 **(d)** 34

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. The average marks of 27 students in a test is 67 and that of 45 students is 35. Find the average marks of all 72 students together. 2

12. A final score combines Maths (score 90, weight 6), Science (score 75, weight 3), English (score 85, weight 3). Find the weighted average score. 2

13. A 100% stacked bar of a school's ₹1700 budget shows 29% on books and 11% on sports. Find the amount on books, on sports and on the remaining items. 2

14. The average marks of 30 students in a test is 53 and that of 27 students is 72. Find the average marks of all 57 students together. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. 110 mL of a 38% sugar solution is mixed with 220 mL of a 44% sugar solution. Find the percentage concentration of the mixture. 3

16. The mean of 10 observations is 53. A new observation 64 is added. Find the new mean. 3

17. The average score of 15 students in a test is 69. When 39 more students take the test, the average score of all 54 students becomes 56. Find the average score of the 39 additional students. 3

OR

390 mL of a 41% sugar solution is mixed with 130 mL of a 13% sugar solution. Find the percentage concentration of the mixture. 3

18. The mean of 27 observations is 30. A new observation 58 is added. Find the new mean. 3

OR

The average score of 24 students in a test is 58. When 18 more students take the test, the average score of all 42 students becomes 55. Find the average score of the 18 additional students. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Three sections have 22, 26 and 38 students with average marks 60, 47 and 44 respectively. Find the average marks of all the students. 5

OR

A family spent ₹2700 in January and ₹4000 in February. Food took 57% of the January total and 49% of the February total (100% stacked bars). In which month was more money spent on food, and by how much? 5

20. Three sections have 40, 44 and 28 students with average marks 43, 78 and 52 respectively. Find the average marks of all the students. 5

OR

A family spent ₹2300 in January and ₹3600 in February. Food took 49% of the January total and 31% of the February total (100% stacked bars). In which month was more money spent on food, and by how much? 5

END OF QUESTION PAPER — Total: 40 Marks